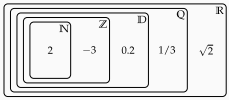
Real numbers and intervals

Lecture and worked examples

Sets of numbers

- $\mathbb{N}$: natural numbers; $\mathbb{Z}$: integers.
- $\mathbb{D}$: decimal numbers, of the form $a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$.
- $\mathbb{Q}$: rational numbers, of the form a/b with $a, b \in \mathbb{Z}$, $b \neq 0$.
- $\mathbb{R}$: real numbers, represented by the abscissae of the points of a graduated line. A real number that is not rational is **irrational**.

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{D} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}.$$



Sets of numbers

Give the smallest of the five sets containing each of the numbers: -95; 7.29; $5/3$; $\sqrt{25}$; $\sqrt{2}$; π; 0.0002.

Worked answer: Sets of numbers

$-95 \in \mathbb{Z}$; $7.29 \in \mathbb{D}$; $5/3 \in \mathbb{Q}$; $\sqrt{25} = 5 \in \mathbb{N}$; $\sqrt{2}, \pi \in \mathbb{R} \setminus \mathbb{Q}$; $0.0002 = 2/10^4 \in \mathbb{D}$. The irrationality of π is accepted here.

Two proofs by contradiction

To prove a statement, assume its negation, deduce a contradiction, then conclude that the statement is true.

Two proofs by contradiction

$\frac{1}{3}$ is not a decimal number.

Two proofs by contradiction

Suppose $1/3 = a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$. Then $10^n = 3a$ would be a multiple of 3. Yet $10 \equiv 1$ modulo 3, so 10^n leaves remainder 1 on division by 3 (the divisibility rule for 3 can also be used). Contradiction. Hence $1/3 \notin \mathbb{D}$.

Two proofs by contradiction

$\sqrt{2}$ is irrational.

Two proofs by contradiction

Suppose $\sqrt{2} = p/q$ with $p, q \in \mathbb{N}$, $q \neq 0$, and p/q in lowest terms. Then $p^2 = 2q^2$, so p^2 is even. If p were odd, its square would be odd: so p is even. Write $p = 2k$. We get $4k^2 = 2q^2$, then $q^2 = 2k^2$; so q is even in turn. Thus 2 divides p and q, contradicting irreducibility. Hence $\sqrt{2} \notin \mathbb{Q}$.

Intervals of the real line

For two real numbers $a < b$, intervals describe the real numbers lying between two endpoints, with or without the endpoints. A square bracket includes the endpoint, a parenthesis excludes it. The symbols $-\infty, +\infty$ are not real numbers and always take a parenthesis.

Intervals of the real line

The strictly negative real numbers form $(-\infty, 0)$: 0 is excluded. $[a, a] = \{a\}$; $(a, a) = \varnothing$.

Intervals of the real line

Write as an interval then represent: $2 \leq x \leq 5$; $x > 4$; $-6 < x \leq 2$; $x < 0$.

Worked answer: Intervals of the real line

$[2, 5]$; $(4, +\infty)$; $(-6, 2]$; $(-\infty, 0)$. On the number line, mark the endpoints, then show each one as included or excluded.

Intersection and union of intervals

$I \cap J$ corresponds to the condition "belonging to I **and** to J"; $I \cup J$ corresponds to "belonging to I **or** to J". A union of intervals is not always an interval.

Intersection and union of intervals

With $I = [0, 12]$ and $J = [3, 20]$, determine $I \cap J$ and $I \cup J$.

Worked answer: Intersection and union of intervals

$I \cap J = [3, 12]$ and $I \cup J = [0, 20]$.

Intersection and union of intervals

Complete, justifying with representations on a number line.

Worked answer: Intersection and union of intervals

For each intersection, keep the points common to both representations; for each union, keep all the points of at least one. In the last row, 0 belongs to neither interval: the union cannot be replaced by $(-2, 3]$.

Absolute value and distance

$|x|$ is the distance from the real number x to 0; $|x| \geq 0$ and

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

The distance between a and b is $|b - a|$. For $a \in \mathbb{R}$ and $r \geq 0$:

$$|x - a| \leq r \iff x \in [a - r, a + r], \qquad |x - a| < r \iff x \in (a - r, a + r).$$

Absolute value and distance

Calculate $|3.2|$, $|-1.6|$, $|\sqrt{3}-2|$, then the distances between 2 and 5, 2 and -5, -2 and -5.

Worked answer: Absolute value and distance

3.2; 1.6; $2 - \sqrt{3}$ since $\sqrt{3} < 2$. The distances are 3, 7 and 3.

Absolute value and distance

Solve: $|x - 7/3| = 14$; $|x - 2| \leq 3$; $|x + 3| < 5$.

Worked answer: Absolute value and distance

For the first equation, $x - 7/3 = 14$ or $x - 7/3 = -14$, so $S = \{-35/3, 49/3\}$. Then $|x - 2| \leq 3 \iff x \in [-1, 5]$. Finally $|x + 3| = |x - (-3)| < 5 \iff x \in (-8, 2)$.

Absolute value and distance

Write $x \in [4, 10]$ as an inequality with an absolute value.

Worked answer: Absolute value and distance

The centre is $(4 + 10)/2 = 7$ and the radius $(10 - 4)/2 = 3$: $|x - 7| \leq 3$.

Bounds and approximations

To bound x is to give $a \leq x \leq b$; the width is $b - a$. An approximation from below is less than or equal to x; an approximation from above is greater than or equal to it. Rounding chooses the nearest value at the required precision. Significant figures are counted from the first nonzero digit; the displayed precision must suit the context.

Bounds and approximations

Bound $\sqrt{2}$ between two consecutive hundredths. Give its rounding to the nearest hundredth.
Round 0.004567 to three significant figures, then 12345 to three significant figures in scientific notation.

Worked answer: Bounds and approximations

$1.41^2 = 1.9881 < 2 < 2.0164 = 1.42^2$, so $1.41 < \sqrt{2} < 1.42$ (width 0.01). The rounding to the nearest hundredth is 1.41. The other roundings are 0.00457 and 1.23×10^4.

Python

```
def encadre_racine2(p):
    # p entier naturel ; calculs entiers exacts
    q = 10**p
    k = 0
    while (k+1)**2 <= 2*q*q:
        k = k+1
    return k/q, (k+1)/q
```

Bounds and approximations

Explain why this function bounds $\sqrt{2}$. What does it return for $p = 2$? Why does it terminate?

Worked answer: Bounds and approximations

The loop looks for the largest integer k such that $k^2 \leq 2q^2$. On exit, $k^2 \leq 2q^2 < (k+1)^2$, so $k/q \leq \sqrt{2} < (k+1)/q$. For $p = 2$, it returns $(1.41, 1.42)$. Since $(2q)^2 > 2q^2$, the counter eventually reaches a value that makes the test false. The final divisions are floating-point approximations; the square tests are exact.